

5 The nervous system enables an organism to respond to environmental stimuli.

(a) Several types of neurone are found in the nervous system.

(i) Which term describes the space between a sensory neurone and a relay neurone?

(1)

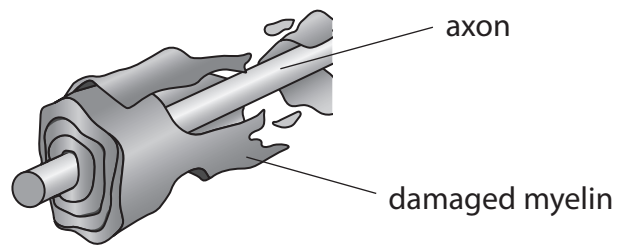
- ☐ **A** axon
- ☐ **B** cell body
- ☐ **C** postsynaptic membrane
- ☐ **D** synaptic cleft

(ii) Compare and contrast the function of a sensory neurone and a motor neurone.

(3)

(b) Guillain-Barré syndrome is a disorder of the nervous system.

The immune system damages the myelin sheath surrounding the axons of individuals with Guillain-Barré syndrome.



Symptoms of this disorder include:

- inability to maintain steadiness when walking
- inability to feel any sensations in the limbs
- loss of reflexes such as knee jerk.

Suggest why a person with Guillain-Barré syndrome will suffer from the symptoms of this condition.

(3)

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(c) Mutations in the genes coding for voltage-gated calcium ion channels have been linked to neurological conditions.

Explain how these mutations might cause these neurological conditions.

(4)

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(Total for Question 5 = 11 marks)